

WILHELMINADORP, Netherlands

WMO: 063230

Lat: 51.5260N

Lon: 3.8836E

Elev: 3

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.3	26.4	16.4	12.6	25.9	20.6	15.5	27.7	35.2	49.1	31.9	48.3	10.2	80	0.596

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.0	82.3	67.7	78.0	65.9	74.6	64.4	69.4	78.8	67.5	75.4	65.9	72.4	8.8	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.1	96.2	73.6	64.5	91.0	71.2	63.2	86.7	69.4	33.5	78.7	31.9	75.6	30.6	72.5	77.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.9	24.2	21.1	DB	18.3	90.0	5.3	4.8	14.6	93.4	11.5	96.3	8.5	99.0	4.7	102.5	(4)
(5)			WB	17.6	73.6	5.0	1.9	13.9	75.0	11.0	76.1	8.2	77.1	4.5	78.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	51.7	39.0	40.1	44.2	50.6	56.3	60.8	64.2	64.4	60.2	53.5	46.2	40.5	(6)
(7)		DBStd	10.76	6.88	6.46	5.93	6.27	5.68	5.10	5.01	4.36	4.37	5.49	5.37	6.63	(7)
(8)		HDD50	1350	342	279	192	68	9	0	0	0	0	28	133	299	(8)
(9)		HDD65	4975	805	697	644	433	275	146	76	64	156	356	563	760	(9)
(10)		CDD50	1983	2	2	13	85	205	324	440	445	305	138	21	3	(10)
(11)		CDD65	136	0	0	0	1	7	20	51	45	11	1	0	0	(11)
(12)		CDH74	881	0	0	0	11	61	150	343	261	51	4	0	0	(12)
(13)		CDH80	214	0	0	0	0	7	40	93	67	7	0	0	0	(13)
(14)	Wind	WSAvg	10.5	12.4	12.0	11.2	10.3	10.1	9.1	8.8	8.7	9.2	10.5	10.8	12.2	(14)
(15)	Precipitation	PrecAvg	33.8	2.8	2.5	1.8	1.6	2.2	2.8	3.2	3.3	3.2	3.2	3.7	3.4	(15)
(16)		PrecMax	44.0	6.7	4.9	4.7	3.2	4.4	5.3	6.8	6.5	8.2	7.1	5.8	6.8	(16)
(17)		PrecMin	23.9	0.2	0.7	0.5	0.0	0.7	0.9	0.6	0.6	0.7	0.6	0.4	0.7	(17)
(18)		PrecStd	4.6	1.5	1.1	1.0	0.9	1.1	1.3	1.7	1.8	2.0	1.5	1.3	1.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.8	57.0	63.4	75.5	80.2	85.4	87.3	86.6	79.2	72.3	61.1	56.0	(19)
(20)			MCWB	52.2	50.4	53.2	58.6	65.2	69.3	69.1	69.2	67.0	62.9	56.9	53.7	(20)
(21)		2%	DB	52.5	53.6	58.8	70.0	74.6	78.1	82.2	80.6	74.3	65.8	58.0	54.0	(21)
(22)			MCWB	50.5	49.7	51.9	57.5	61.7	66.4	67.4	68.1	64.7	60.2	55.0	51.8	(22)
(23)		5%	DB	50.2	51.3	55.1	65.3	70.4	73.3	77.6	76.1	70.5	63.1	55.8	51.8	(23)
(24)			MCWB	48.2	48.4	50.6	55.2	60.1	64.1	66.4	65.9	63.0	59.1	53.3	49.8	(24)
(25)		10%	DB	48.2	49.0	52.6	60.6	66.4	69.9	73.8	72.4	67.3	61.1	53.6	49.6	(25)
(26)			MCWB	46.5	46.7	49.0	53.1	58.5	62.7	64.9	64.1	61.0	57.7	51.2	47.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.8	52.3	55.4	61.3	68.2	71.5	71.7	71.9	68.3	63.7	57.4	54.4	(27)
(28)			MCD B	54.3	55.0	60.4	71.4	77.2	81.4	83.2	82.3	76.7	69.8	59.7	55.4	(28)
(29)		2%	WB	50.7	50.4	53.1	58.2	63.8	67.4	69.3	69.1	65.8	61.2	55.4	52.1	(29)
(30)			MCD B	52.3	52.6	56.9	67.5	70.4	75.7	78.6	78.0	72.0	64.8	57.6	53.6	(30)
(31)		5%	WB	48.4	48.6	51.2	55.9	61.1	65.2	67.3	67.1	63.9	59.7	53.4	50.0	(31)
(32)			MCD B	49.8	50.7	54.4	64.0	68.6	71.8	75.4	73.9	68.7	62.6	55.6	51.6	(32)
(33)		10%	WB	46.6	46.8	49.3	53.8	58.9	63.3	65.6	65.2	62.4	58.1	51.5	47.8	(33)
(34)			MCD B	48.1	48.8	52.3	59.6	65.4	68.9	72.4	70.7	66.7	60.7	53.3	49.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.0	8.7	11.1	14.2	13.9	14.1	14.0	14.0	12.5	10.4	8.0	7.0	(35)
(36)			MCDBR	8.2	11.7	15.6	21.7	20.9	20.5	21.1	20.2	17.4	12.5	9.3	8.1	(36)
(37)			MCWBR	7.7	8.9	10.2	10.9	10.8	11.5	9.9	9.5	9.2	8.3	7.8	7.7	(37)
(38)		5% WB	MCDBR	8.2	10.1	13.6	19.4	19.5	18.4	18.5	17.7	15.2	11.8	9.3	8.2	(38)
(39)			MCWBR	7.9	8.4	9.7	10.5	11.3	11.1	9.9	9.5	9.1	8.3	8.2	8.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.336	0.356	0.391	0.418	0.426	0.422	0.427	0.418	0.395	0.372	0.349	0.332	(40)	
(41)		taud	2.360	2.343	2.252	2.212	2.238	2.269	2.263	2.306	2.359	2.402	2.375	2.339	(41)	
(42)		Ebn at Noon	211	240	252	259	263	265	261	257	249	232	203	190	(42)	
(43)		Edh at Noon	20	27	35	41	42	41	40	37	31	25	20	18	(43)	
(44)	All-Sky Solar Radiation	RadAvg	236	457	844	1378	1632	1759	1656	1416	1036	594	276	179	(44)	
(45)		RadStd	23	84	109	128	120	134	178	117	76	49	27	21	(45)	

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days				
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (84 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page